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Theoretical investigation of the excitation of leaky modes for multilayered models

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SUMMARY

The dispersion analysis with array-based methods enables us to detect and analyse the multimode dispersive waves with weak amplitudes. Over the past years, there has been widespread extraction of higher normal modes from earthquake records and ambient noise, which have been extensively employed in near-surface and crustal investigations. Notably, the frequent reporting of leaky modes in recent observations has gained significant attention. This is because leaky modes can provide more constraints on subsurface structures. Particularly, a subset of leaky modes is found to be sensitive to *P*-wave velocities, which has the potential to compensate for the limitations of traditional surface wave inversion. Nevertheless, due to the lack of knowledge of the excitation and practical application of leaky modes, it is urgent to have an effective guide in selecting certain leaky modes for practical inversion combined with other imaging methods. To this end, the theoretical investigation into the excitation of leaky modes for multilayered models is undertaken. By computing theoretical seismograms for various source mechanisms and source depths with the discrete wavenumber and the normal-mode summation methods, respectively, the qualitative evaluation of the leaky-mode contributions to the synthesized seismograms is conducted. Additionally, the impact of the source depth for the same source mechanism is meticulously examined by the dispersion spectra. Furthermore, with the accurate computation of leaky modes, we categorize the leaky modes into PL and OP (organ-pipe) modes according to their distinctive properties, that is, their attenuation and eigen-displacements, which may be used to explain very well the occurrence of certain leaky modes on the dispersion spectrum and be indicative of the reasonable choice of PL modes to put them into practical inversion for *P*-velocity structures.

Key words: Guided waves; Surface waves and free oscillations; Theoretical seismology.

1 INTRODUCTION

The surface wave dispersion plays an important role in the imaging of near-surface and crustal *S*-wave velocity structures. The fundamental modes of surface waves have been widely used in the dispersion curve inversion because of its high amplitudes and dominance in the seismograms. With the development of the array-based dispersion analysis methods (e.g. Frequency–Bessel method, Wang *et al.* 2019; Li *et al.* 2020) that enable us to measure the dispersion of multiple-mode waves with low amplitudes, there has been a transition from solely using the fundamental mode to incorporating higher modes in the surface wave dispersion inversion. In addition to surface waves, a series of recent publications highlighted the successful extraction of leaky modes that travel faster than surface waves but typically exhibit lower amplitudes in seismograms and noise cross-correlation functions (e.g. Li *et al.* 2021, 2022; Wang *et al.* 2024). Due to the potential of leaky modes in estimating subsurface structures, a great interest in leaky modes arises and consequent studies have been undertaken both from theoretical and observational or application perspectives (Gao *et al.* 2014; García-Jerez & Sánchez-Sesma 2015; Znak *et al.* 2015; Wu & Chen 2017; Furumura & Kennett 2018; Ma *et al.* 2019; Li *et al.* 2021; Li *et al.* 2022).

Leaky modes physically describe the waves that leak at least *S* waves into the infinite substratum when propagating in an open waveguide. Since the PL wave (Sommville 1930), a long-period (> 10 s) wave train beginning shortly after the initial *P* wave, was first identified with the fundamental leaky mode (Oliver & Major 1960), subsequent observations of wave trains which can be interpreted in terms of leaky modes had been reported (e.g. Oliver 1961, 1964;

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Gilbert & Laster 1962; Su & Dorman 1965). During the same stage, the theory of leaky modes had also experienced a series of developments (e.g. Rosenbaum 1960; Phinney 1961; Gilbert 1964; Watson 1972), and later the computation of complete synthetic seismograms by mode summation was also completed (Haddon 1984, 1986, 1987).

Leaky modes have long been suggested to constrain the *P*-wave velocity structure, which may offer unique advantages to situations where traditional surface waves and classic seismic reflection and refraction methods may fall short (e.g. Oliver & Major 1960; Su & Dorman 1965; Roth *et al.* 1998). In the traditional surface wave inversion which typically only retrieves the *S*-wave velocity, the *P*-wave velocity is often determined empirically. Constraining the subsurface structures using normal and leaky modes are potential to provide richer structural information including *P*- and *S*-wave velocities, and Poisson's ratios, which serve as important references for studying the lithology, rheological state and other characteristics of subsurface layers (e.g. Nakajima *et al.* 2001; Zhang *et al.* 2009, 2024; Wang *et al.* 2024). From the perspective of inversion, although the traditional surface wave inversion is not sensitive to the *P*-wave velocity, excessive inaccuracy in *P*-wave velocities can still adversely affect the inversion of *S*-wave velocities due to the crosstalks between elastic parameters. Consequently, the simultaneous inversion of *P*- and *S*-wave velocities leads to better convergence in inversion and resultantly more accurate models (Li *et al.* 2022).

On the way towards the practical inversion using leaky modes, earlier studies had been devoted to the effects of the velocity structures on the leaky modes (e.g. Su & Dorman 1965; Haskell 1966; De Bremaecker 1967; Cochran *et al.* 1970; Stalmach & De Bremaecker 1973) and to the attenuation and the excitation of the leaky modes (e.g. Laster *et al.* 1965; Haskell 1966; Dainty 1971). They found that a deeper event in the crust could more efficiently excite the fundamental leaky mode (the most commonly seen mode) than a near-surface source (Dainty 1971). Notwithstanding the advancements after 1960, leaky modes have found limited applications in imaging crustal structures (Su & Dorman 1965; Ibrahim 1969; Fujita & Nishimura 1991). In the past decades, the application of leaky modes was prevalent in near-surface seismic exploration (Roth *et al.* 1998; Roth & Holliger 1999; Boiero *et al.* 2013; Gao *et al.* 2014). However, some studies approximately compute leaky modes using acoustic theory due to the high Poisson's ratios of typical near-surface structures (Shtivelman 2004; Park *et al.* 2005) while exact computation and analysis with elastic theories are rarely reported.

Given the reported observations and applications of leaky modes, accurate modal computation may be a limiting factor in incorporating leaky modes into practical inversion. Furthermore, it is found in the published papers that the computed leaky modes significantly outnumber their actual appearance in the dispersion spectra (Roth *et al.* 1998; Wu & Chen 2017; Li *et al.* 2021). Then two questions need to be answered in this study: (i) what kind of conditions may favour the excitation of the leaky modes, which requires considering the effects of the source and the media properties but restricts herein to the study of the former factor only; and (ii) why massive leaky modes are not observed in the dispersion spectrum, which may be explained by considering the attenuation of leaky modes as well as their eigen-displacements. By answering these two questions, we wish to establish a solid theoretical foundation for realistic velocity inversion using leaky modes. The outline of this paper is as follows: we first introduce the basic theories of leaky modes and make two inferences on the leaky modes based on the formulae for independently synthesizing seismograms and then introduce the method and the strategy of accurately computing leaky modes in Section 2; the contribution of leaky modes to the seismograms and the effects of the source on the excitation of leaky modes are illustrated in Section 3, and the dispersive calculations of leaky modes together with the attenuation and the eigen-displacements of leaky modes are also shown for two cases in this section, with a careful analysis being performed to explain the resultant theoretical dispersion spectrum; finally, summarizing the main results of our investigation, a detailed conclusion is given in the concluding section.

2 THEORY AND METHODS

2.1 Basis of leaky modes

In this section, we briefly introduce the theory of leaky modes based on the residue theorem given a lack of related exposition in recent literature. First, we revisit the formulae of seismic responses that are expressed with the integral along the real wavenumber (denoted by k) axis. Following Zhou & Zhang (2020), the Green's function of the seismic waves in horizontal stratified models can be calculated using

$$G(r,\theta,z;\omega)=\frac{1}{2}\sum_{m}\int_{-\infty}^{+\infty}F_m(\theta)\frac{A_m(k,\omega,z)}{S(k,\omega)}H_m^{(2)}(kr)\,\mathrm{d}k, \tag{1}$$

where $F_m(\theta)$ is related to the source and the azimuth, $A_m(k,\omega,z)$ represents a smooth, non-singular function which does not depend on the source mechanism but includes only the source depth, $S(k,\omega)$ denotes the secular function (i.e. the dispersion function), $\frac{A_m(k,\omega,z)}{S(k,\omega)}$ is also called the kernel function, $H_m^{(2)}(\cdot)$ denotes the Hankel function of the second kind, r denotes the horizontal distance and m denotes the order of the Hankel function. In terms of the real frequency and pure elastic media, two branch points and several zero points lie on the real-k axis (Fig. 1), which renders the integral in eq. (1) improper. Zeros of the secular function (i.e. poles of the integrand function) are also called dispersion points, each of which denotes an individual mode. The branch points for the *P–SV* problem, which are $k=\omega/\alpha_{N+1}$ and $k=\omega/\beta_{N+1}$, donate that the vertical wavenumbers of P and S waves in the half-space, that is,$\gamma^{(N+1)}=\sqrt{k^2-(\omega/\alpha^{(N+1)})^2}$ and $\nu^{(N+1)}=\sqrt{k^2-(\omega/\beta^{(N+1)})^2}$, are zero, where N is the number of layers above the half-space, α and β represent *P*- and *S*-wave velocities, respectively.

Since the straightforward integration in eq. (1) seems numerically difficult, the residue theorem is used to analyse the Green's function (Rosenbaum 1960). Considering the duality of the vertical wavenumbers, we may define four Riemann sheets according to the signs of the real parts of the multivalued functions $\gamma^{(N+1)}$ and $\nu^{(N+1)}$ (Watson 1972). The Riemann sheet for which $\mathrm{Re}\{\gamma^{(N+1)}\}>0$ and $\mathrm{Re}\{\nu^{(N+1)}\}<0$ is designated as the $(+,-)$ sheet, and similarly the $(-,-)$ sheet means $\mathrm{Re}\{\gamma^{(N+1)}\}<0$ and $\mathrm{Re}\{\nu^{(N+1)}\}<0$. These sheets are divided by the Sommerfeld cuts which donate the curves meeting $\mathrm{Re}\{\gamma^{(N+1)}\}=0$ and $\mathrm{Re}\{\nu^{(N+1)}\}=0$. Typically, for real frequencies and pure elastic media, Sommerfeld cuts lie on the real-k axis exactly (Fig. 1). Among the Riemann sheets, only the $(+,+)$ sheet is called the proper sheet since modes on this sheet satisfy the radiation boundary condition, that is, the corresponding eigen wavefields tend to zero at infinity. The other

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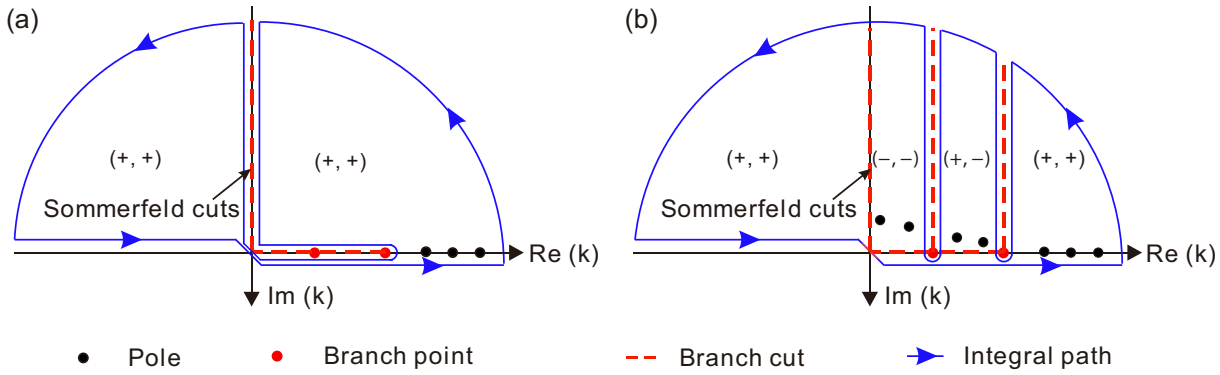


Figure 1. The integration contour using different branch cuts when calculating the *P–SV* wavefield. (a) Using Sommerfeld cuts and (b) vertical branch cuts.

sheets break the radiation boundary condition in the half-space (*cf.* Appendix A).

For the integrand in eq. (1), possible branch cuts can be arbitrary lines that start from the branch point and extend to the infinity. Since the integral along the great circle tends to zero, the Green's function can be calculated with the residues of discrete modes inside the contour and the integral along the branch cuts (Fig. 1). When Sommerfeld cuts are chosen as the branch cuts (Fig. 1a), the complete wave solution can be decomposed into the residues of the modes on the (+, +) sheet (i.e. normal modes), and the additional integrals along Sommerfeld cuts. Typically, summing all the normal-mode contributions produces surface waves straightforwardly. Meanwhile, the integrals along the Sommerfeld cuts represent leaky waves.

When the Sommerfeld cuts are chosen as the branch cuts, there is no room for leaky modes. To expose the leaky modes, the integral contour is specifically designed by using vertical branch cuts (Jensen *et al.* 2011), as shown in Fig. 1(b). Then, the leaky waves are contributed by modes located on other sheets and the integrals along vertical branch cuts. Once the integral contour crosses the Sommerfeld cuts, the function should change to another Riemann sheet. Therefore, all the modes located on (+, +), (+, −) and (−, −) sheets contribute to the Green's function (Fig. 1b). Note that the (−, +) sheet is not included in the integral path and thus deserves no attention. The modes located on (+, −) and (−, −) sheets belong to leaky modes and should be considered for full wavefields. Generally, leaky modes can approximate the leaky wave inside the waveguide quite well and have been manifested as an effective tool for the investigation of leaky-wave dispersion and attenuation. Since the significant contribution to the waveform comes mainly from the (+, −) sheet (Wang & Herrmann 1980), carefully examined in our study are only the leaky modes in this sheet, which have phase velocities in the range of $[\beta^{(N+1)}, \alpha^{(N+1)}]$ and describe travelling waves leaking only *S*-wave energy into the substratum (Haddon 1984).

2.2 Waveform simulation

Directly simulating the leaky waves requires the determination of complex dispersion points and the integral along branch cuts, which requires sophisticated mathematical consideration. Therefore, we evaluate the contributions of leaky modes based on two mature synthetic methods: the discrete wavenumber method (DWM; Bouchon & Aki 1977; Chen & Zhang 2001; Zhang *et al.* 2003), by which the generated seismogram is complete, and the normal-mode summation method (NMM; Aki & Richards 1980) for synthesizing surface waves solely. The DWM confronts the challenge of handling the improper integral in eq. (1). The classical approach considers the complex frequency, which forces the dispersion and branch points to deviate from the real axis. To obtain the correct seismograms, the effects of complex frequencies are compensated after the real-k integral is completed (Chen 1999).

Following Aki & Richards (1980), the NMM expresses the displacement of surface waves due to a point source with the summation related to the secular functions and normal-mode eigenfunctions. For example, the Rayleigh waves excited by a point vertical force can be written as

$$\mathbf{u}^{\text{Rayleigh}}(r,\theta,z,t) = e^{-i\omega t}\sum_m\sum_n \frac{f_R(k_n,m)r_2(h)}{8cUI} \left[r_1(z)\mathbf{S}_{k_n}^{m\,(1)}(r,\theta) + r_2(z)\mathbf{R}_{k_n}^{m\,(1)}(r,\theta)\right], \quad (2)$$

where $\mathbf{R}_k^m(r,\theta)$ and $\mathbf{S}_k^m(r,\theta)$ are two of the vector basis functions in the cylindrical system and are written as $\mathbf{R}_k^{m\,(1)}(r,\theta)$ and $\mathbf{S}_k^{m\,(1)}(r,\theta)$ with the m-order first-kind Hankel function, n is the index of the normal modes, r_1 and r_2 denotes the horizontal and vertical eigen-displacements of a Rayleigh-wave mode, respectively, f_R the source term, h and z the depth of the source and of the receiver, and the definitions of the phase velocity c, the group velocity U and the energy integral I are referred to Aki & Richards (1980).

By scrutinizing eqs (1) and (2), we can make two inferences. First, the dispersion spectrum (Wang *et al.* 2019; Li & Chen 2020) generated from the kernel function in eq. (1) may correlate with the modal intensity calculated by eq. (2), although the latter relies also on the source mechanism. Second, the contribution of leaky modes is just the difference of the seismograms synthesized by the two methods. Experimental evidence will be given in Section 3 to show the validity of the above two inferences. It should be noted that the second inference is based on the fact that the time response contributed by the branch-cut integral vanishes after the first arrival (Haddon 1986, 1987) and amounts to nothing in the seismogram.

2.3 Determining the leaky modes on the complex-wavenumber plane

Calculating the eigen wavenumbers of leaky modes for multilayered models is accomplished in the same manner as that of normal modes of surface waves. The only difference is that leaky modes are usually complex roots of the dispersion equation while normal modes are real roots. When assuming the

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real-complex approach (Watson 1972), the roots of leaky modes are sought in the complex wavenumber domain for each given frequency.

Based on the generalized reflection/transmission coefficients method (Chen 1993; Chen & Chen 2002; He *et al.* 2006; Wu & Chen 2016, 2017, 2022), the dispersion equation for solving the leaky modes, which shares an identical form with that for the normal modes, is obtained as

$$\det\left\{\widehat{\mathbf{R}}_{\mathrm{ud}}^{j-1}\widehat{\mathbf{R}}_{\mathrm{du}}^{j}-\mathbf{I}\right\}=0(j=1,2,\ldots,N), \tag{3}$$

where $\widehat{\mathbf{R}}_{\mathrm{ud}}^{j-1}$ and $\widehat{\mathbf{R}}_{\mathrm{du}}^{j}$ are the generalized reflection coefficients associated with the upper and the lower interfaceS in the jth layer, and $\mathbf{I}$ is the identity matrix. Note that the left-hand side of eq. (3) is a family of secular functions (He *et al.* 2006; Wu & Chen 2016). By selecting different values for j, we can get different secular functions. Theoretically, all the secular functions have the same zeros. However, diverse numerical efforts may be required to locate a specific mode using different secular functions (Wu & Chen 2016). A thorough search of normal modes for models with fluid layers or abnormal properties may necessarily invoke several secular functions. However, one suitable dispersion equation may be adequate to find the complex roots of leaky modes (Wu & Chen 2017).

The direct strategy for searching roots is the grid search in the 2-D complex plane, but it becomes time-consuming and yields less accurate results (Roth *et al.* 1998; Gao *et al.* 2014). Alternatively, some iterative methods such as the Newton–Raphson method can be employed. With a series of initial values, the roots may be located very efficiently (Gilbert 1964; Cochran *et al.* 1970; Watson 1972; Radovich & De Bremaecker 1974). Similar to the approach of Watson (1972), we search the roots from high to low frequency, get the trial value of the root of the same order at the next frequency point by extrapolation, and employ the Newton–Raphson method to achieve sufficiently accurate roots. Once all of the leaky roots within the dictated frequency range are found and a check for the mode missing is conducted, we may then proceed to compute the eigenfunctions of the leaky modes required, in the same manner as that of the normal modes (Wu & Chen 2016, 2022). Numerical examples will be given for two cases in Section 3.

3 RESULTS AND ANALYSIS

In this section, numerical tests are performed to analyse the excitation of leaky modes for different models. Since leaky modes attract great attention in both the crustal and near-surface investigations, we select three typical models including a crust model, a near-surface and a shallow-water models.

3.1 A crustal model

3.1.1 Contribution of leaky modes to the seismograms

We compute the theoretical seismograms using the DWM and NMM, respectively, for the crust-upper-mantle model (Fig. 2) modified from Shen & Ritzwoller (2016). Consider the two source types shown in Table 1, and the Green solutions are computed at a distance of 500 km from the focus of 0.5 and 8 km depth, as shown in Figs 3 and 4, respectively. In both figures, the results for the strike-slip source are labelled as (a), and those for the dip-slip source are labelled as (b); each subplot shows the difference between the results

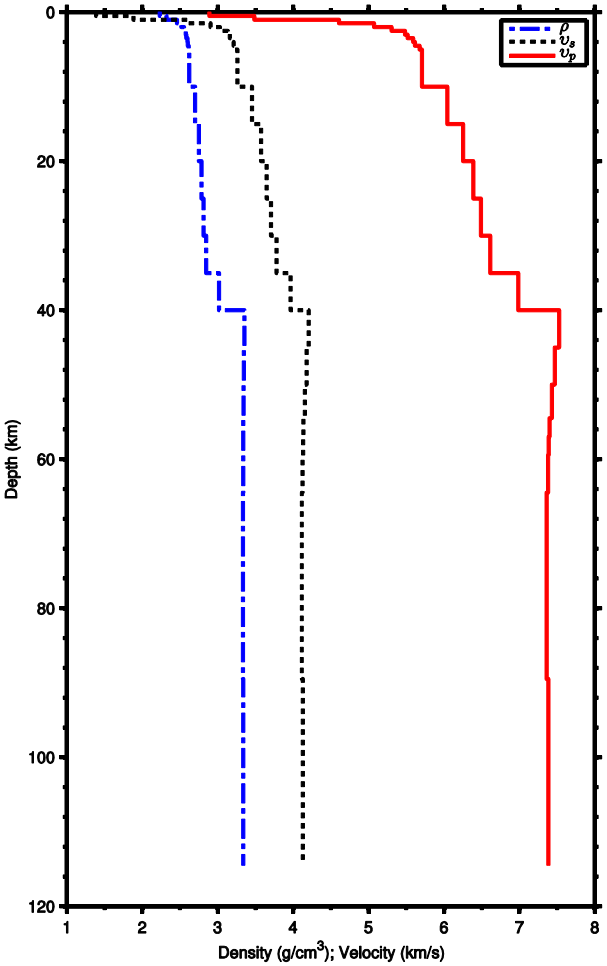


Figure 2. The crust-upper-mantle model modified from Shen & Ritzwoller (2016).

Table 1 Source parameters.

Fault type	Rake (°)	Dip (°)	Strike (°)	Azimuth (°)
Strike slip	180	90	45	30
Dip slip	90	45	45	30

of DWM and NMM. Because the NMM can only synthesize surface waves but the DWM produces complete seismograms including the early-arrival body waves (see the signals between the onset of P and S waves in Fig. 4), this difference is attributed to the leaky waves.

As shown in Fig. 3, when the source depth is 0.5 km, the generated synthetic seismograms from the two methods are almost the same, whether for the strike-slip source or for the dip-slip. For a shallow source, the seismograms observed at the surface are mainly controlled by surface waves, while the leaky waves are quite weak and are almost imperceptible in the complete seismograms. Fig. 4 shows that, however, when the source is 8 km deep, the differences between the seismograms computed from the two methods become dramatic for both source types, especially for the strike-slip one.

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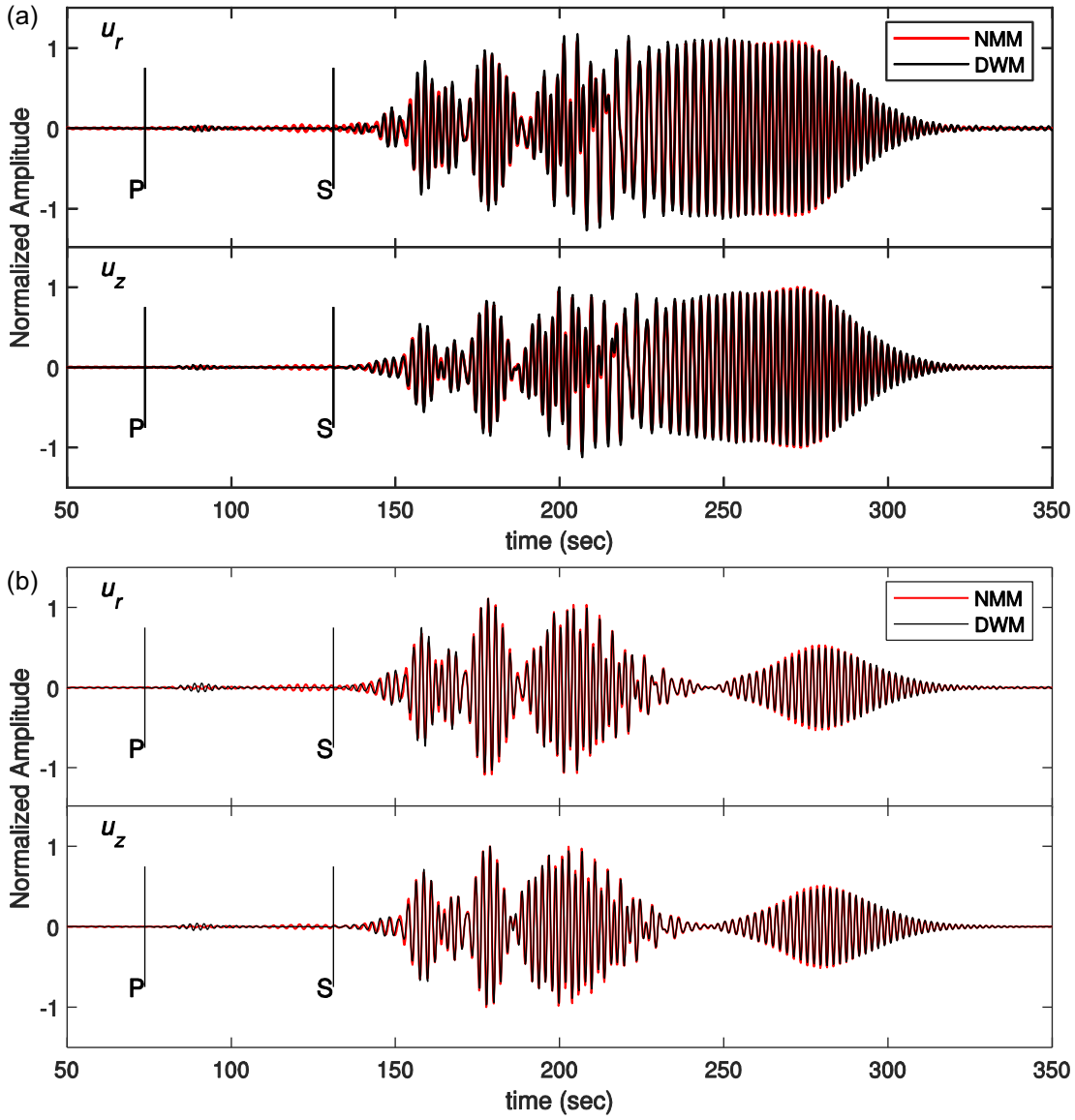


Figure 3. The synthetic seismograms (the radial and the vertical components) for the seismic depth 0.5 km for the strike-slip (a) and the dip-slip (b) sources. The results computed by the NMM and DWM are marked by red and black lines, respectively. The onset of *P* and *S* waves is also indicated by vertical lines in each panel. The epicentral distance is 500 km. The Hanning taper is applied to 0.01–0.55 Hz.

The results of Figs 3 and 4 indicate that the deeper focus could effectively trigger the PL phase on the seismogram, as also observed by Dainty (1971).

3.1.2 Effects of the source on the excitation of leaky modes

It can also be demonstrated using corresponding dispersion spectra that the deep focus favours the excitation of leaky modes. As shown in Fig. 5, more higher order leaky modes appear on the dispersion spectrum for the source depth of 8 km than those for the source depth of 0.5 km. Note here that the dispersion spectra computed from the kernel function are blind to the difference between various source mechanisms and depend only on the source depth, the frequency and the wavenumber, as the receiver is fixed on the surface.

We then proceed to study the effects of the source on the excitation of leaky modes in view of the mode intensity. Since the excited strength of normal modes could be easily obtained from eq. (2), the effects of the source on the excitation of normal modes are assumed to have similar effects on the excitation of leaky modes, as will be verified later. Fig. 6 gives diagrams of the excited strength of the normal modes for the model shown in Fig. 2 for the two types of sources in Table 1 with different source depths. We infer from the results for normal modes that, when the focus depth is shallow (Figs 6a and c), the source mechanism has a prominent effect on the excitation of leaky modes, and when the source is deep (Figs 6b and d), the effect of the source mechanism seems immaterial. Further, for the same source mechanism, the source depth still has a significant effect on the excitation of leaky modes, which is consistent with the results of Helmberger & Engen (1980).

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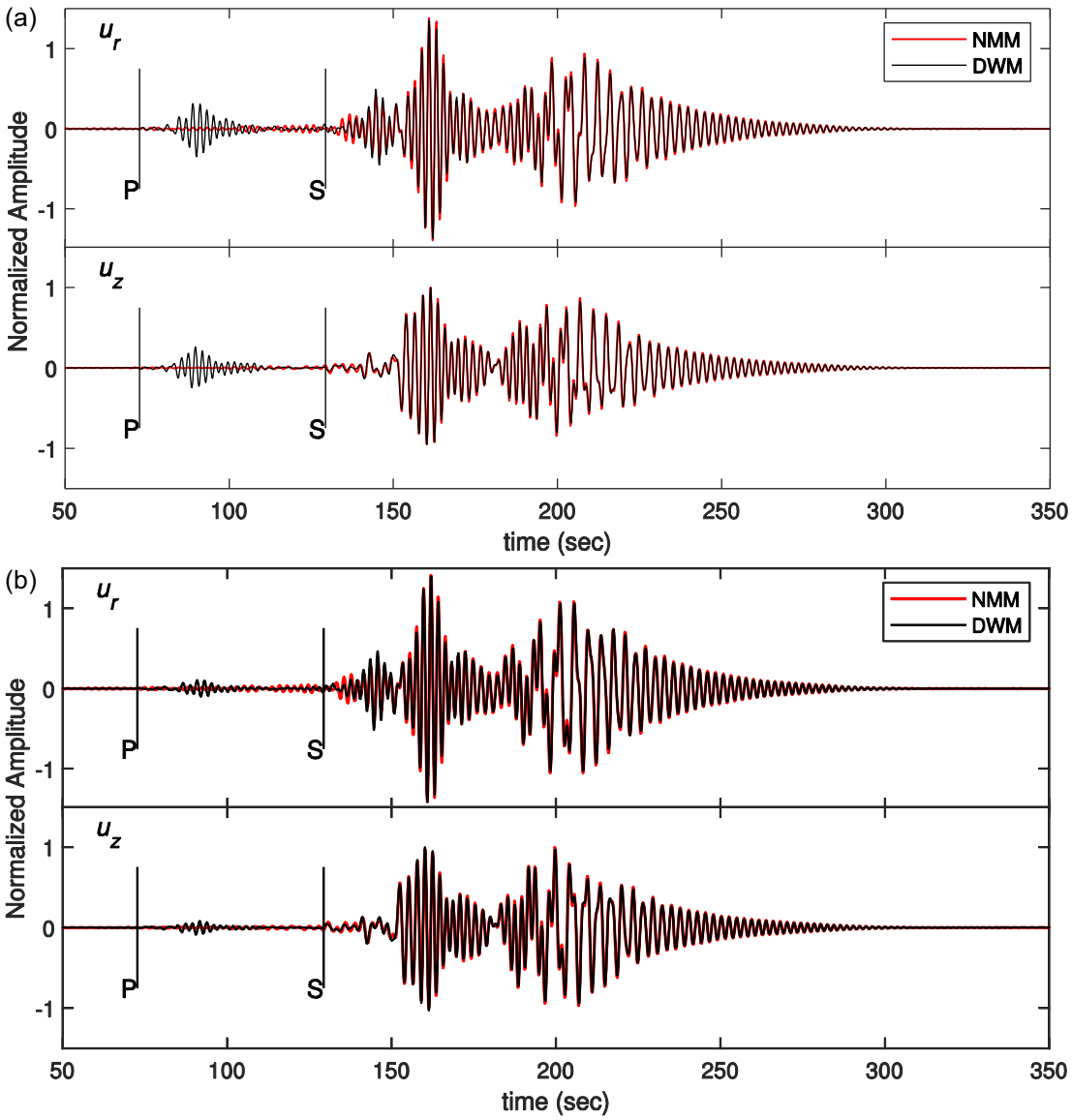


Figure 4. Same as Fig. 3, except that the source depth is 8 km. Note that the differences between the waveforms produced by the DWM and the NMM between the onset of *P* and *S* waves show the contribution of the leaky modes.

3.2 A near-surface model

We further use the model shown in Table 2 proposed by Roth & Holliger (1999) to study the leaky modes observed in near-surface prospecting. The model is composed of a homogeneous layer and an underlying half-space, which represents that the sediments are separated by a strong velocity contrast (Roth *et al.* 1998). Fig. 7 shows the dispersion spectra for different source depths with the receiver fixed on the free surface. It can be clearly revealed from Fig. 7 that, with the increasing source depth, the intensity of the excited leaky modes becomes stronger, which once again confirms that a deeper seismic event may favour the excitation of leaky modes. Besides the dispersion spectrum, we also calculate the theoretical dispersion curves using the method described in eq. (3). The frequency range is from 10^{-5} to 200 Hz, and the frequency spacing is 1 Hz. The dispersion curves (including normal and leaky modes) are compared with the dispersion spectra (see the upper-left panel in Fig. 7). The computed normal and leaky modes are in good agreement with the dispersion spectrum plotted from the kernel function. However, it is evident that some leaky modes do not appear on the spectrum, which warrants a detailed investigation of the behaviour of leaky modes.

Adopting the notation of Watson (1972), the leaky modes computed in the $(+,\ -)$ Riemann sheet are categorized into two classes: the organ-pipe modes (abbreviated as OP modes) of relatively steep slope and the PL modes of gentle ramp; the latter could be approximated by acoustic normal modes as shown by blue open circles in Fig. 8. The acoustic normal modes are obtained by considering a purely fluid models whose *P*-wave velocities are identical with the original elastic models and imposing the traction-free condition at the surface and the radiation boundary condition in the half-space. It is noteworthy that due to the essentially different origins of the two types of leaky modes, their phase-velocity curves can freely

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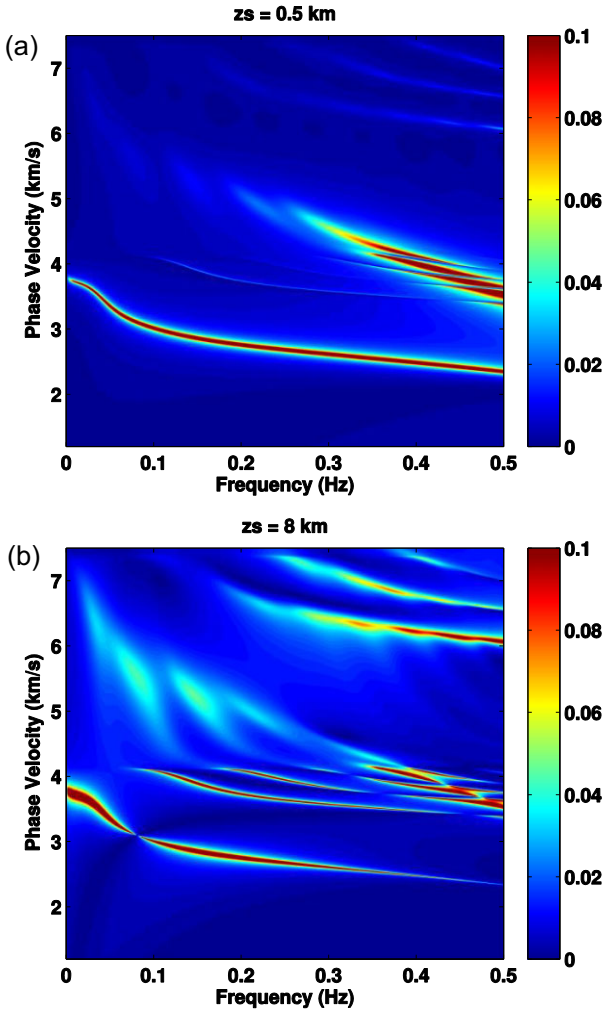


Figure 5. The theoretical dispersion spectra computed from the kernel function: (a) source depth $z_s = 0.5$ km and (b) source depth $z_s = 8$ km.

Table 2 A near-surface model, taken from Roth & Holliger (1999).

Layer no.	Thickness (m)	ρ (g cm^{-3})	β (m s^{-1})	α (m s^{-1})
1	10	1.6	330	1100
2	∞	2.0	540	1800

intersect each other when assuming real frequency, the modes being connected by lines based on numerical order (Fig. 8). Thus it is a distinct pattern of the leaky-mode dispersion contrasted with the well-known avoided-crossing character of the normal-mode dispersion curves (Wu & Chen 2016).

Let us examine then the oscillating character of the two types of modes. We plot the eigen-displacements of PL modes (Fig. 9) and OP modes (Fig. 10) in the rectangle in Fig. 8. Because the eigen-displacements of leaky modes are complex due to their complex horizontal wavenumbers, only the real parts of the eigen-displacements are displayed, which will not be explicitly stated hereafter. We see from Fig. 9 that the energy of the PL modes is mainly concentrated in the region of the 10-m layer thickness and becomes negligibly small as soon as they enter the half-space. Therefore, the oscillating character of the PL modes is very similar to that of normal modes. Without taking the attenuation into account, the PL modes contribute significantly to the seismogram when the source is located in layers above the half-space. On the contrary, the OP modes have little energy in the first layer but their eigen-displacements grow exponentially in the half-space (Appendix A). Consequently, without considering the attenuation, the OP modes are deemed to be difficult to appear on the seismogram when the source is well above the half-space. We mention in passing that, although the PL modes in Fig. 8 could be approximated by the acoustic normal modes, they have essentially different physical mechanisms. The distinctions are evident when comparing the eigen-displacements of the acoustic modes in Fig. 11 with those of the PL modes in Fig. 9.

Next, we consider the attenuation of leaky modes. Fig. 12 shows that with increasing frequency, the attenuation of each order of leaky mode decreases gradually. Thus, the leaky modes of phase velocities just larger than the maximum shear wave velocity tend to appear on the dispersion spectrum, as shown in Fig. 7(a). In particular, the three PL modes in Fig. 12 have extremely small attenuation compared with the rest of the leaky modes, which may account for the more frequent manifestation of the PL modes in near-surface surveys (Roth *et al.* 1998). The attenuation curves of the three PL modes are drawn in Fig. 13. It is evident that the three PL modes all display maxima and minima periodically, which may elucidate the pronounced tuning effect of leaky modes reported in the literature (Roth *et al.* 1998; Znak *et al.* 2015). As shown in Fig. 13, some of the PL modes at certain frequencies may even have extremely small attenuation coefficients (close to zero), which are called slowly attenuating *P–SV* leaky waves and have been extensively studied by García-Jerez & Sánchez-Sesma (2015).

With the analysis presented above, we are aware that the massive OP modes disappear on the dispersion spectra, partly because their attenuations are relatively stronger compared with the PL modes. Most importantly, they contribute little energy just beneath the surface, as can be seen from the variation of their eigen-displacements with depth (Fig. 10). Plotted in Fig. 14 are the eigen-displacements of the leaky modes at 180 Hz where there is distinct appearance of leaky modes when the source depth is 15 m. Judging from the oscillating character of leaky modes we have grasped, with the aid of the dispersion diagram (Fig. 8), we could more definitely determine which are OP modes and which correspond to PL modes, and explain why some leaky modes are visible on the dispersion spectrum while the others are not.

3.3 A shallow-water model

Now let us consider a shallow-water model (Table 3), whose maximum *S*-wave velocity is smaller than the *P*-wave velocity in the water. This model is used to study the leaky modes observed in shallow-marine seismic exploration (e.g. Klein *et al.* 2005; Boiero *et al.* 2013; Shi *et al.* 2023). For such a model, due to the observation by Wu *et al.* (2020) that the usual dispersion equation is difficult to find any normal modes, the dispersion equation corresponding to the first solid layer (i.e. setting j in eq. 3 to be 2) is employed to compute the Rayleigh dispersion curves (including normal and leaky modes) shown in Fig. 15. In this case there are many PL modes, which are likewise approximated by the acoustic normal modes denoted by blue dashed lines. Same with the OP modes for the near-surface model (Fig. 8), the OP modes for the shallow-water model are also reverse continuations of the normal modes behind their cut-off frequencies. A significant feature of the PL-mode dispersion curves

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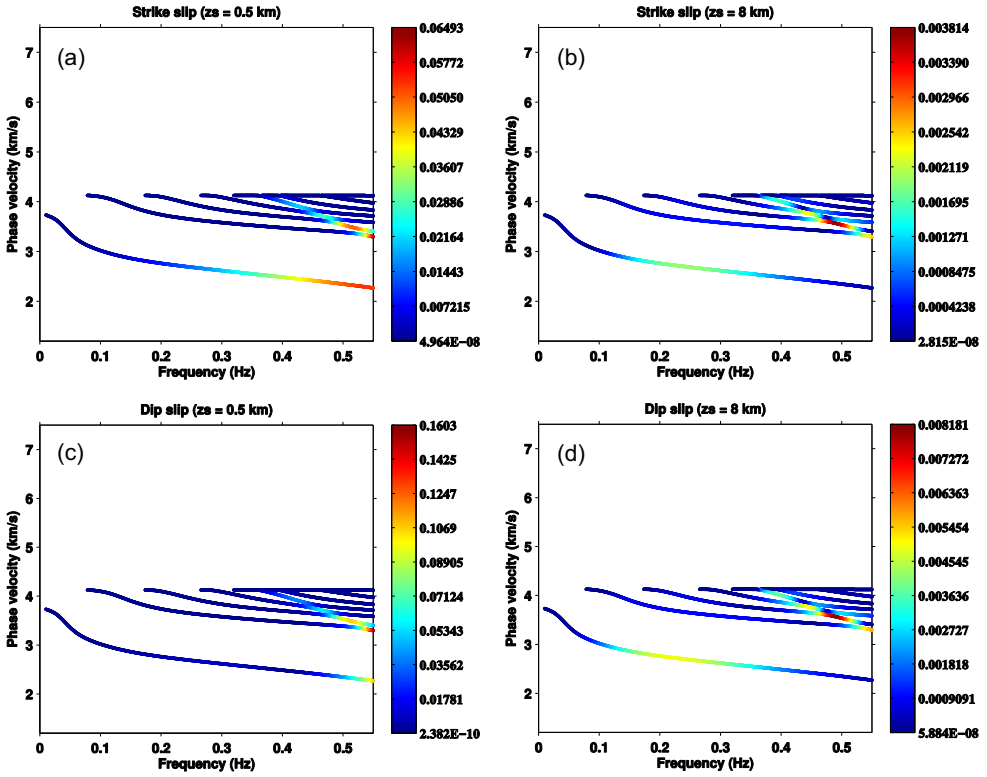


Figure 6. The diagrams of the excited strength of the normal modes for the model shown in Fig. 2: (a) strike-slip source with depth 0.5 km; (b) strike-slip source with depth 8 km; (c) dip-slip source with depth 0.5 km and (d) dip-slip source with depth 8 km.

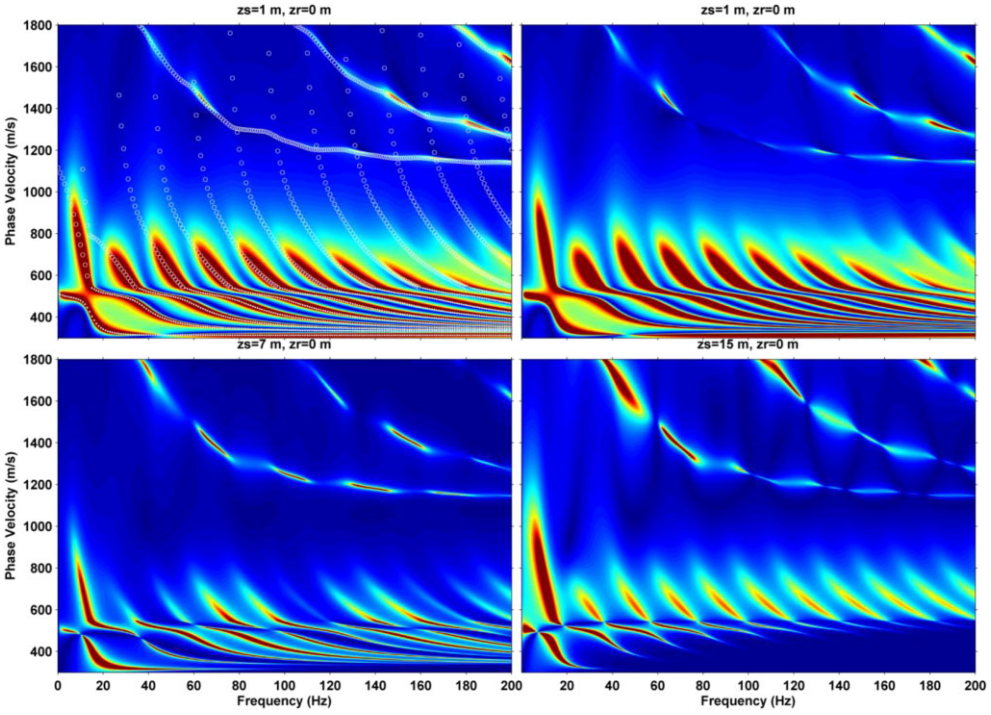


Figure 7. The dispersion spectrum computed for the near-surface model (Table 2). The corresponding source depth and receiver depth are indicated above each subplot. The normal modes (solid dots) and leaky modes (open circles) are superimposed on the first subplot for reference.

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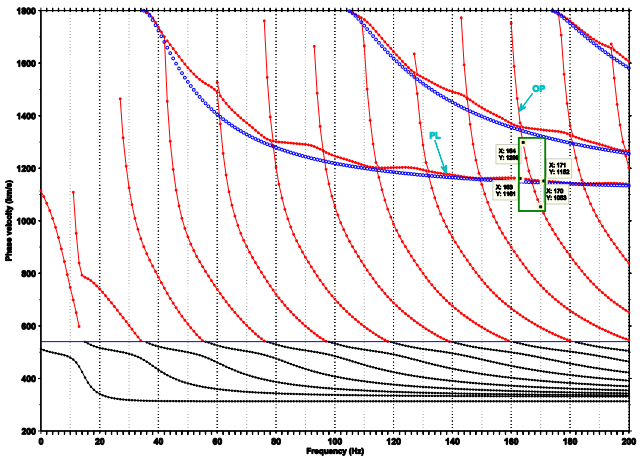


Figure 8. Dispersion curves of normal modes (black dots) and leaky modes (red dots) for the model shown in Table 2. Two distinct families of modes—PL and OP modes—are separately indicated. The horizontal blue line represents the maximum shear wave velocity. The acoustic normal modes (blue open circles) are superimposed for comparison with PL modes. The eigen-displacements of the leaky modes in the green rectangle would then be calculated and plotted.

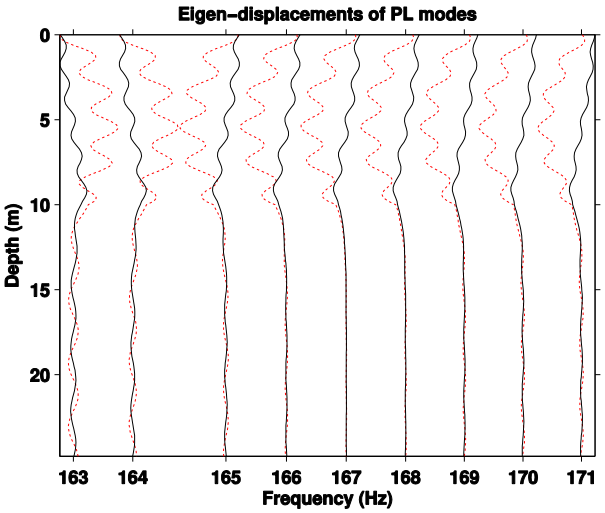


Figure 9. The eigen-displacements (real parts) of PL modes in the rectangular box in Fig. 8. The vertical and horizontal displacements are denoted by black solid line and red dotted line, respectively.

is the occurrence of the positive slopes (e.g. around 40 Hz of the second PL mode in Fig. 15). For normal nodes, the appearance of positive slopes usually relies on the existence of low-velocity layers. However, our test shows that the leaky modes can exhibit positive-slope dispersion without any low-velocity layers. As shown in Appendix B, PL modes located in the region of positive-slope dispersion exhibit quite large group velocities, stronger attenuation (than other PL modes of the same order), and are more dominant with *S* waves. However, a rigorous physical explanation requires further study.

Fig. 16 is the attenuation diagram of the leaky modes in Fig. 15. Once again, with the increasing frequency the attenuations of the leaky modes of the same order gradually decrease, and the attenuations of the PL modes are particularly small. Therefore, judging

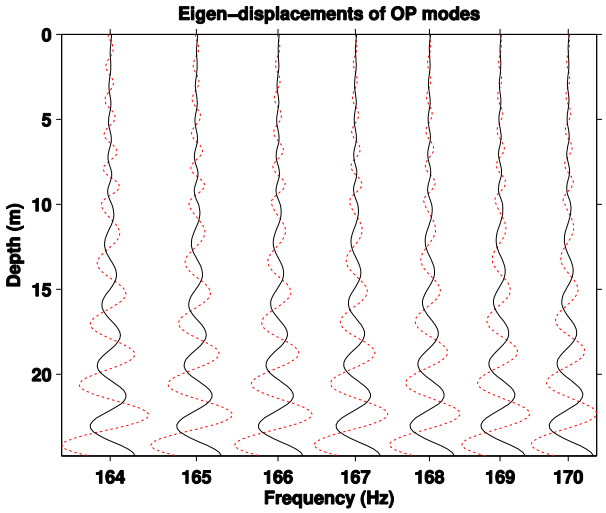


Figure 10. The eigen-displacements (real parts) of OP modes in the rectangular box in Fig. 8 (the same denotation of line types and colour as Fig. 9).

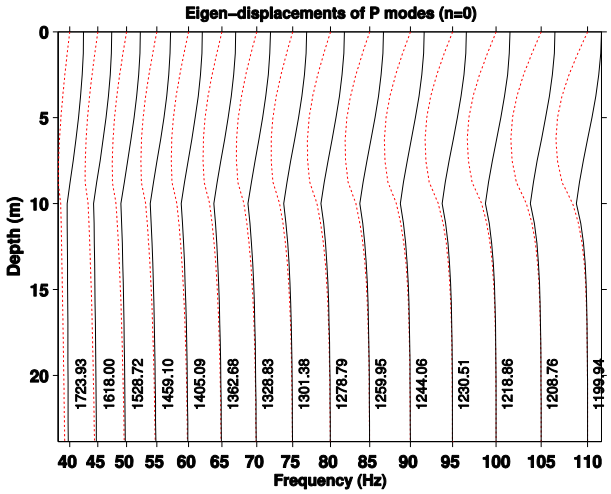


Figure 11. The eigen-displacements of the fundamental ($n = 0$) acoustic normal modes (*P* modes) at selected frequencies for the model shown in Table 2, but with vanishing values of shear velocities. The phase velocity of each mode is labelled appropriately. The denotation of line types and colours is the same as Fig. 9.

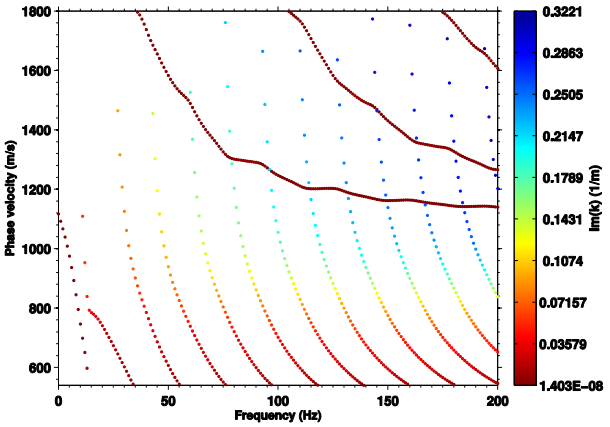


Figure 12. The attenuation of the leaky modes shown in Fig. 8.

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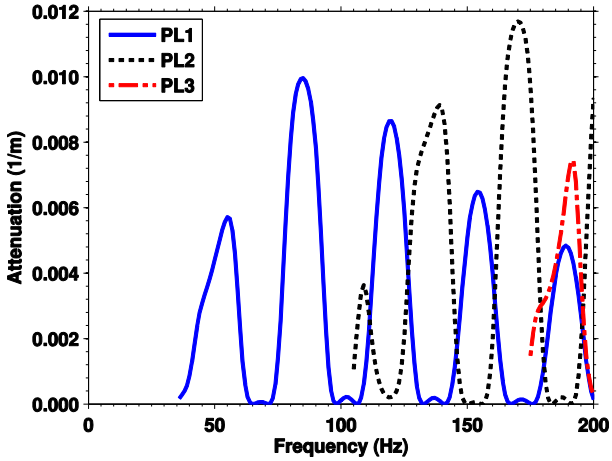


Figure 13. The attenuation curves of the three PL modes shown in Fig. 8.

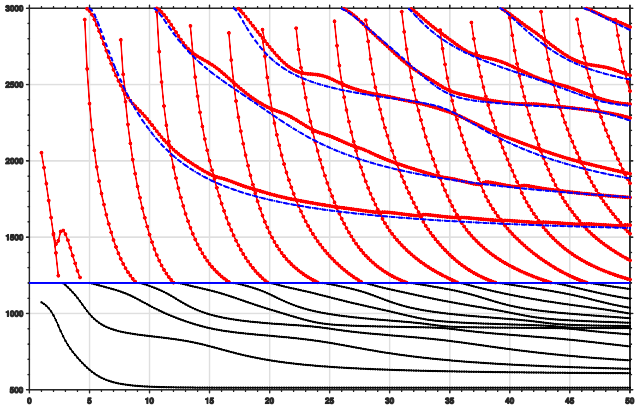


Figure 15. Dispersion curves of normal modes (black dots) and leaky modes (red dots) for the model shown in Table 3. The acoustic normal modes (blue dashed lines) are superimposed for comparison with PL modes.

Table 3 A shallow-water model.

Layer no.	Thickness (m)	ρ (g cm^{-3})	β (m s^{-1})	α (m s^{-1})
1	40	1.0	0	1500
2	40	1.5	600	1700
3	100	1.8	900	2300
4	∞	2.0	1200	3000

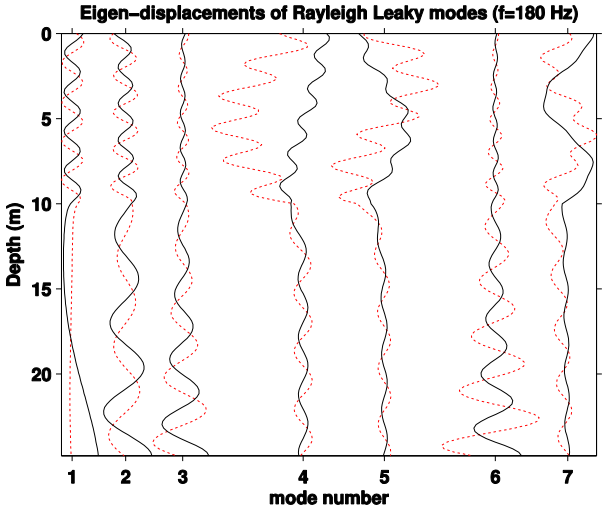


Figure 14. The eigen-displacements (real parts) of the leaky modes for the near-surface model (Table 2) at 180 Hz (the same denotation of line types and colour as Fig. 9).

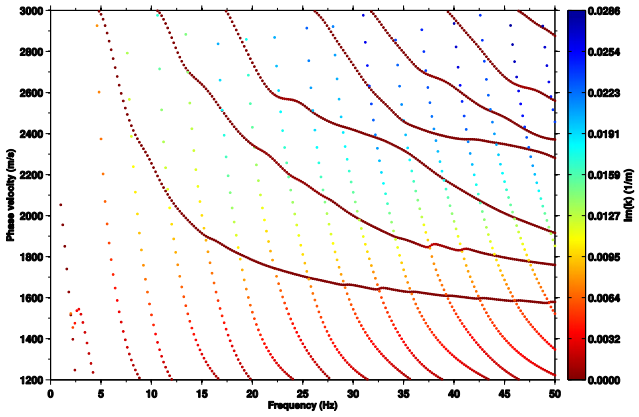


Figure 16. The attenuation of the leaky modes shown in Fig. 1.

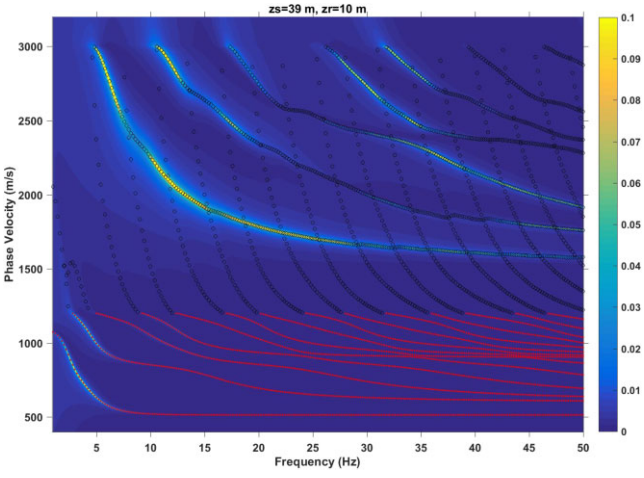


Figure 17. The dispersion spectrum computed for the shallow-water model (Table 3). The corresponding source depth and receiver depth are labelled above. The normal modes (solid dots) and leaky modes (open circles) are superimposed for reference.

from the attenuation diagram alone, we could predict that the leaky modes whose phase velocities are slightly larger than the maximum *S*-wave velocity and the PL modes are apt to appear on the dispersion spectrum. Nevertheless, we should bear in mind that, the excitation of modes also depends upon the source mechanism and the source–receiver configuration.

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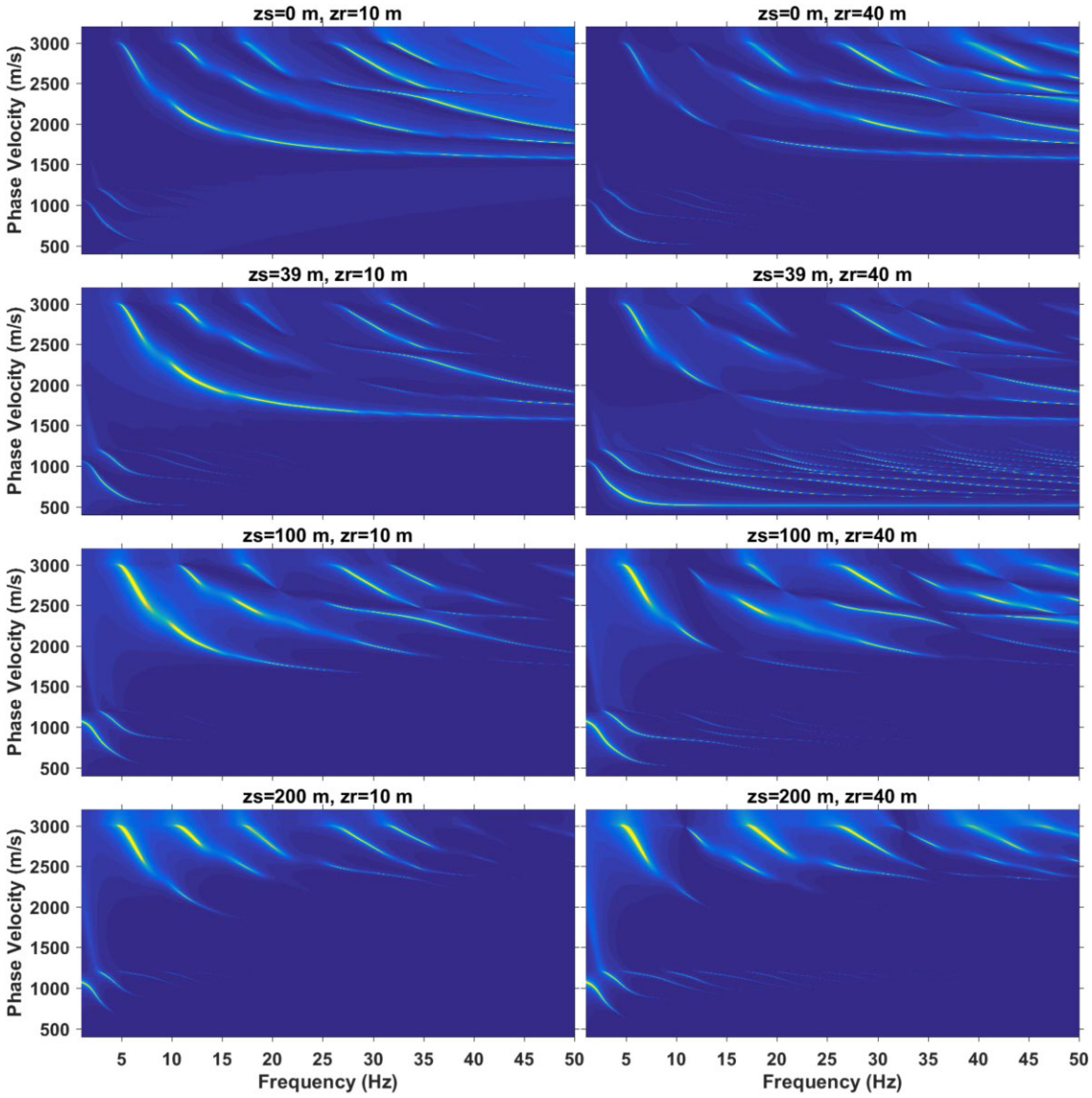


Figure 18. The dispersion spectrum for the shallow-water model (Table 3) at different source depths (0, 39, 100 and 200 m) and receiver depths (10 and 40 m).

Fig. 17 shows the dispersion spectrum when the source is just above the seafloor and the receiver is only 10 m beneath the surface. Superimposed on the spectrum are the normal and leaky modes, represented by solid dots and open circles, respectively. The bright regions above the maximum shear wave velocity agree very well with the computed PL leaky modes. We further notice that the energy of the leaky modes on the dispersion spectrum is evidently stronger than that of the normal modes. This situation does not change much with different source–receiver configurations (Fig. 18) except when the receiver is just located on the seafloor, in which case a number of normal modes appear but are still excited less strongly on the dispersion spectrum. This phenomenon may be well accounted for by the concept of 'adaptive mode observers' proposed by Wu & Chen (2016); one that is located at the depth where the eigen-displacements of a certain excited mode are significant, tends to feel the energy of this mode more strongly. Moreover, for a shallow-water model whose maximum shear velocity is still lower than the sound velocity in the water, the energies of all of the normal modes are trapped mainly in the seafloor (Wu *et al.* 2020).

The dispersion spectra with different source–receiver configurations are displayed in Fig. 18. All the receiver depths are 10 m on the left panel and 40 m on the right panel, and the source depth ranges are from 0 to 200 m. As a whole, with the increase of the source depth leaky modes are excited more strongly, but when the source is too deep (note the case when the source is 200 m deep), the number of the excited leaky modes decreases instead. The excited modes are the most when the receiver is closer to the source, as just discussed and well understood.

The eigen-displacements of the leaky modes are selectively computed at 31 Hz (Fig. 19). Again, the oscillating character of the OP modes is distinct from that of the PL modes; the former have little energy in the water, while the latter have significant energy in the upper region of the model. The discontinuity of the horizontal displacement across the ocean floor is seen for all of the modes. Because the PL modes in Fig. 19 are found to have

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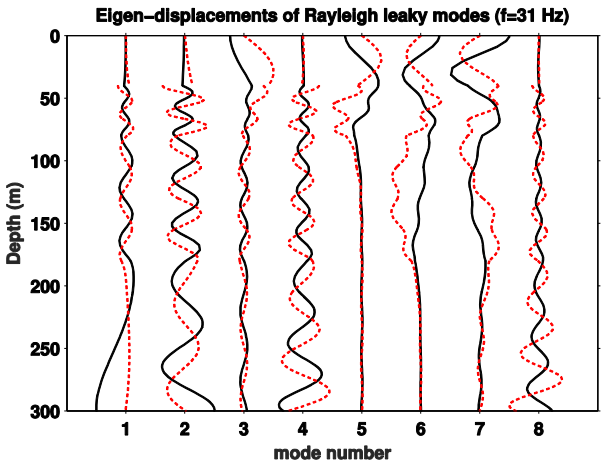


Figure 19. The eigen-displacements (real parts) of the leaky modes for the shallow-water model (Table 3) at 31 Hz (the same denotation of line types and colour as Fig. 9).

negligible energy below 200 m deep, this may account for the decrease in the number and the strength of the excited modes when the source depth is increased to 200 m (Fig. 18). Were the source not an explosive source but a horizontal force, some OP modes would be expected to appear on the spectrum, as illustrated by Znak *et al.* (2015).

4 CONCLUSIONS

Motivated by the two questions: (i) under what conditions the leaky modes are favourably excited, and (ii) why many theoretically computed leaky modes are not present in the dispersion spectrum, we have investigated the excitation of leaky modes for multilayered models in view of their attenuation and eigen-displacements. Based on the synthetic seismograms generated by the DWM and the NMM, we find that for a given model the contribution of leaky modes to the resultant seismogram depends on the source mechanism and the source depth (the receiver is assumed to be on the surface). The effect of the source mechanism on the excitation of leaky modes is minimal for deep sources, whereas this effect may be prominent for shallow ones. Moreover, for the same source mechanism, the source depth may play an important role in the excitation of leaky modes. Therefore, deeper events tend to favour the excitation of leaky modes, while for shallow sources different source mechanisms contribute much.

Accurate computation of leaky modes is performed in the $(+,\ -)$ Riemann sheet. The leaky modes can be classified into two types: PL and OP modes. PL modes generally have little attenuation and have a significant contribution to the early portion of the seismograms. PL modes are more sensitive to *P*-wave velocities than to *S*-wave velocities. In contrast, OP modes have larger attenuation but have a tendency to lower attenuation with the increasing frequency. Moreover, OP modes are more sensitive to shear wave velocities and are reverse continuations of normal modes beyond their cut-off frequencies. These results may help explain why frequently present on the dispersion spectrum above the maximum shear wave velocity are the PL modes and partial OP modes whose phase velocities are slightly larger than the maximum shear wave velocity. Most OP modes are absent on the dispersion spectrum because these modes generally attenuate more heavily compared to the PL modes, and also because the OP modes have little energy in the shallower subsurface regions in view of their eigen-displacements relative to those of the PL modes. However, certain source mechanisms may still excite some OP modes for shallow sources. Furthermore, the periodic appearance of the maximas and minimas of the attenuation curves of PL modes are held accountable for the pronounced tuning effect in the dispersion spectrum.

Judging from the attenuation and the eigen-displacements of leaky modes, we are greatly aided to analyse the dispersion spectrum. With the recent soaring interest in applying an array-based method to extract leaky-mode dispersion curves and in an attempt to invert the *P*-wave velocities (Li *et al.* 2021, 2022; Wang *et al.* 2024), one may be further served by this study to suitably select the PL leaky modes combined with normal modes to perform joint inversion to obtain reliable *P*-wave velocity structure.

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AUTHOR CONTRIBUTIONS

Bo Wu wrote the manuscript, did the formulae derivation, and performed most of the calculations; Caiwang Shi validated the results and improved the manuscript; Xiaofei Chen provided guidance for this work, and Qi Zhang helped polish the text.

DATA AVAILABILITY

No field data were used in this study. All of the model data used are explicitly shown or cited appropriately in the text. The computer codes of the DWM for synthesizing complete seismogram can be accessed at https://zenodo.org/records/13852674 and the code package 'mode_sum_PSV' for synthesizing Rayleigh waves by mode summation can be obtained from https://zenodo.org/records/13852670. The input files and the simulated results are also available in the two uploaded code packages. The code package for the theoretical dispersion spectrum can be accessed at https://zenodo.org/records/4574722. The results of dispersion curves and the associated eigenfunctions where needed in this study were produced by the versatile solver of the normal modes for horizontal stratified complicated models, which can be downloaded from the public repository link https://github.com/BranchInVine/SurfaceModes. The code of the computation of leaky modes will be shared on reasonable request

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to the corresponding author. All the figures were ploted using MATLAB (available at https://www.mathworks.com/products/matlab.html) except that Fig. 1 was drawn using CorelDRAW (available at https://www.coreldraw.com/en/). All the websites were last accessed in September 2024.

REFERENCES

Aki, K. & Richards, P.G., 1980. *Quantitative Seismology: Theory and Methods,* W. H. Freeman, San Francisco.

Alsop, L.E., Goodman, A.S. & Gregersen, S., 1974. Reflection and transmission of inhomogeneous waves with particular application to Rayleigh waves, *Bull. seism. Soc. Am.,* **64,** 1635–1652.

Boiero, D., Wiarda, E. & Vermeer, P., 2013. Surface- and guided-wave inversion for near-surface modeling in land and shallow marine seismic data, *Leading Edge,* **32,** 638–646.

Bouchon, M. & Aki, K., 1977. Discrete wavenumber representation of seismic source wave fields, *Bull. seism. Soc. Am.,* **67,** 259–277.

Chen, W. & Chen, X., 2002. Modal solutions in stratified multilayered fluid-solid half-space, *Sci. China Ser. D—Earth Sci.,* **45,** 358–365.

Chen, X.-f. & Zhang, H.-m., 2001. An efficient method for computing Green's functions for a layered half-space at large epicentral distances, *Bull. seism. Soc. Am.,* **91,** 858–869.

Chen, X., 1993. A systematic and efficient method of computing normal mode for multilayered half space, *Geophys. J. Int.,* **115,** 391–409.

Chen, X., 1999. Seismogram synthesis in multi-layered half-space part I. Theoretical formulations, *Earthq. Res. China,* **13,** 149–174.

Cochran, M.D., Woeber, A.F. & De Bremaecker, J.C., 1970. Body waves as normal and leaking modes, 3. Pseudo modes and partial derivatives on the (+−) sheet, *Rev. Geophys. Space Phys.,* **8,** 321–357.

Dainty, A.M., 1971. Leaking modes in a crust with a surface layer, *Bull. seism. Soc. Am.,* **61,** 93–107.

De Bremaecker, J.C., 1967. Body waves as normal and leaking modes, part I: introduction, *Bull. seism. Soc. Am.,* **57,** 191–198.

Fujita, M. & Nishimura, K., 1991. Estimation of crustal structure from observed group velocities of PL-waves in Japan, *Pure appl. Geophys,* **137,** 233–249.

Furumura, T. & Kennett, B.L.N., 2018. Regional distance PL phase in the crustal waveguide—An analog to the teleseismic W phase in the upper-mantle waveguide, *J. geophys. Res.: Solid Earth,* **123,** 4007–4024.

Gao, L., Xia, J. & Pan, Y., 2014. Misidentification caused by leaky surface wave in high-frequency surface wave method, *Geophys. J. Int.,* **199,** 1452–1462.

García-Jerez, A. & Sánchez-Sesma, F.J., 2015. Slowly-attenuating P-SV leaky waves in a layered elastic halfspace. Effects on the coherences of diffuse wavefields, *Wave Motion,* **54,** 43–57.

Gilbert, F. & Laster, S.J., 1962. Experimental investigation of PL modes in a single layer, *Bull. seism. Soc. Am.,* **52,** 59–66.

Gilbert, F., 1964. Propagation of transient leaking modes in a stratified elastic waveguide, *Rev. Geophys.,* **2,** 123–153.

Haddon, R.A.W., 1984. Computation of synthetic seismograms in layered earth models using leaking modes, *Bull. seism. Soc. Am.,* **74,** 1225–1248.

Haddon, R.A.W., 1986. Exact evaluation of the response of a layered elastic medium to an explosive point source using leaking modes, *Bull. seism. Soc. Am.,* **76,** 1755–1775.

Haddon, R.A.W., 1987. Response of an oceanic wave guide to an explosive point source using leaking modes, *Bull. seism. Soc. Am.,* **77,** 1804–1822.

Haskell, N.A., 1966. The leakage attenuation of continental crustal P waves, *J. geophys. Res.,* **71,** 3955–3967.

He, Y., Chen, W. & Chen, X., 2006. Normal mode computation by the generalized reflection-transmission coefficient method in planar layered half space, *Chin. J. Geophys.,* **49,** 1074–1081.

Helmberger, D.V. & Engen, G.R. 1980. Modeling the long-period body waves from shallow earthquakes at regional ranges, *Bull. seism. Soc. Am.,* **70**(5), 1699–1714.

Ibrahim, A.-B.K., 1969. Leaking and normal modes as a means to determine crust-upper mantle structure for different paths to Sweden, *Bull. seism. Soc. Am.,* **59,** 1695–1712.

Jensen, F.B., Kuperman, W.A., Porter, M.B., Schmidt, H. & Tolstoy, A., 2011. *Computational Ocean Acoustics,* Springer, New York.

Klein, G., Bohlen, T., Theilen, F., Kugler, S. & Forbriger, T. 2005. Acquisition and inversion of dispersive seismic waves in shallow marine environments, *Mar. Geophys. Res.,* **26,** 287–315.

Laster, S.J., Foreman, J.G. & Linville, F., 1965. Theoretical investigations of modal seismograms for a layer over a half-space, *Geophysics,* **30,** 571–596.

Li, Z. & Chen, X., 2020. An effective method to extract overtones of surface wave from array seismic records of earthquake events, *J. geophys. Res.: Solid Earth,* **125,** e2019JB018511, doi:10.1029/2019JB018511.

Li, Z., Shi, C. & Chen, X., 2021. Constraints on crustal P wave structure with leaking mode dispersion curves, *Geophys. Res. Lett.,* **48,** e2020GL091782, doi:10.1029/2020GL091782.

Li, Z., Shi, C., Ren, H. & Chen, X., 2022. Multiple leaking mode dispersion observations and applications from ambient noise cross-correlation in Oklahoma, *Geophys. Res. Lett.,* **49,** e2021GL096032, doi:10.1029/2021GL096032.

Ma, D.-y., Zhang, Y.-x., Xie, Z.-n., Gao, J.-h. & Yin, J.-W., 2019. Sea-ice thickness measurement based on ice layer waveguide theory, *Tech. Acoust.,* **38,** 479–482.

Nakajima, J., Matsuzawa, T., Hasegawa, A. & Zhao, D. 2001. Three-dimensional structure of vp, vs, and Vp/vs beneath northeastern Japan: implications for arc magmatism and fluids, *J. geophys. Res.: Solid Earth,* **106**(B10), 21843–21857.

Oliver, J. & Major, M., 1960. Leaking modes and the PL phase, *Bull. seism. Soc. Am.,* **50,** 165–180.

Oliver, J., 1961. On the long period character of shear waves, *Bull. seism. Soc. Am.,* **51,** 1–12.

Oliver, J., 1964. Propagation of PL waves across the United States, *Bull. seism. Soc. Am.,* **54,** 151–160.

Park, C.B., Miller, R.D., Xia, J., Ivanov, J., Sonnichsen, G.V., Hunter, J.A. & Christian, H. 2005. Underwater MASW to evaluate stiffness of water-bottom sediments, *Leading Edge,* **24**(7), 724–728.

Phinney, R.A., 1961. Leaking modes in the crustal waveguide, part 1. The oceanic PL wave, *J. geophys. Res.,* **66,** 1445–1469.

Radovich, B.J. & De Bremaecker, J.C., 1974. Body waves as normal and leaking modes—leaking modes of love waves, *Bull. seism. Soc. Am.,* **64,** 301–306.

Rosenbaum, J.H., 1960. The long time response of a layered elastic medium to explosive sound, *J. geophys. Res.,* **65,** 1577–1613.

Roth, M. & Holliger, K., 1999. Inversion of source-generated noise in high-resolution seismic data, *Leading Edge,* **18,** 1402–1406.

Roth, M., Holliger, K. & Green, A.G., 1998. Guided waves in near-surface seismic surveys, *Geophys. Res. Lett.,* **25,** 1071–1074.

Shen, W. & Ritzwoller, M.H., 2016. Crustal and uppermost mantle structure beneath the United States, *J. geophys. Res.: Solid Earth,* **121,** 4306–4342.

Shi, C., Ren, H. & Chen, X., 2023. Dispersion inversion for P-and S-wave velocities based on guided P and Scholte waves. *Geophysics,* **88(** 6), R721–R736.

Shtivelman, V. 2004. Estimating seismic velocities below the sea-bed using surface waves, *Near Surf. Geophys.,* **2**(4), 241–247.

Sommville, O., 1930. A propos d'une onde longue dans la première phase de quelques séismogrammes, *Gerlands Beitr. Geophys.* , **27,** 437–442.

Stalmach, D.M. & De Bremaecker, J.C., 1973. Body waves as normal and leaking modes: dispersion and excitation on the (+−) sheet, *Bull. seism. Soc. Am.,* **63,** 995–1011.

Su, S.S. & Dorman, J., 1965. The use of leaking modes in seismogram interpretation and in studies of crust-mantle structure, *Bull. seism. Soc. Am.,* **55,** 989–1021.

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Wang, C.Y. & Herrmann, R.B., 1980. A numerical study of P-, SV-, and SH-wave generation in a plane layered medium, *Bull. seism. Soc. Am.,* **70,** 1015–1036.

Wang, J., Wu, G. & Chen, X., 2019. Frequency-bessel transform method for effective imaging of higher-mode Rayleigh dispersion curves from ambient seismic noise data, *J. geophys. Res.: Solid Earth,* **124,** 3708–3723.

Wang, Z., Sun, C. & Wu, D., 2024. Simultaneous estimation of P-and S-wave velocities by integrated inversion of guided-P and surface wave dispersion curves, *Surv. Geophys.,* **45,** 429–458.

Watson, T.H., 1972. A real frequency, complex wavenumber analysis of leaking modes, *Bull. seism. Soc. Am.,* **62,** 369–384.

Wu, B. & Chen, X., 2016. Stable, accurate and efficient computation of normal modes for horizontal stratified models, *Geophys. J. Int.,* **206,** 1281–1300.

Wu, B. & Chen, X., 2017. Accurate computation of leaky modes for anomalous layered models, *Ann. Geophys.,* **60,** S0663, doi:10.4401/ag-7477.

Wu, B. & Chen, X., 2022. A versatile solver of the normal modes for horizontal stratified complicated models, *Seismol. Res. Lett.,* **93,** 1852–1867.

Wu, B., Li, Z. & Chen, X., 2020. Some aspects of dispersion calculation for models with surficial fluid layers, in *Annual Meeting of Chinese Geoscience Union (CGU),* CGU, Beijing Botong Digital Publishing House, Chongqing, p. 4477, doi:10.26914/c.cnkihy.2020.058617.

Zhang, H.-M., Chen, X.-F. & Chang, S., 2003. An efficient numerical method for computing synthetic seismograms for a layered half-space with sources and receivers at close or same depths, *Pure appl. Geophys.,* **160,** 467–486.

Zhang, H., Thurber, C. & Bedrosian, P. 2009. Joint inversion for vp, vs, and vp/vs at SAFOD, Parkfield, California, *Geochem. Geophy. Geosy.,* **10**(11), doi:10.1029/2009GC002709.

Zhang, Z., Nakata, N., Karplus, M., Kaip, G., Qin, L., Li, Z. & Chen, X., 2024. Seismic full-wavefield imaging of the West Antarctic Ice Sheet interior near the ice flow divide, *Earth planet. Sci. Lett.,* **636,** 118701, doi:10.1016/j.epsl.2024.118701.

Zhou, J. & Zhang, H. 2020. Exact Rayleigh-wave solution from a point source in homogeneous elastic half-space, *Bull. seism. Soc. Am.,* **110**(2), 783–792.

Znak, P., Kashtan, B. & Bakulin, A., 2015. Guided waves and Rayleigh leaking modes with outpost algorithm, in *77th EAGE Conference and Exhibition 2015,* pp. 1–5, European Association of Geoscientists & Engineers, IFEMA Madrid, Spain.

APPENDIX A:THE GROWTH OF THE LEAKY-MODE EIGENFUNCTIONS

It is well known that the eigenfunctions of normal modes decrease exponentially with depth in the half-space, satisfying the radiation boundary conditions. However, as mentioned in Section 3.3, the leaky-mode eigenfunctions show exponential growth in the half-space, which does not meet the radiation boundary conditions. This interesting feature of leaky modes had been recognized once the leaky-mode theory was developed. Many physical, theoretical and numerical works have been proposed to explain the exponential growth (e.g. Gilbert 1964; Alosp *et al.* 1974; Haddon 1984; Jensen 2011). Considering that the leaky modes discussed in this study are located on the (+, −) sheet, the growth of their eigenfunctions can be simply explained mathematically with the analytical expressions of the corresponding *P*- and *S*-wave potentials in the half-space:

$$p^{(N+1)}(z) = p^{(N+1)}(z_N)\exp\left[\gamma^{(N+1)}(z - z_N)\right] \qquad (z \geq z_N), \quad \text{(A1)}$$

$$s^{(N+1)}(z) = s^{(N+1)}(z_N)\exp\left[\nu^{(N+1)}(z - z_N)\right] \qquad (z \geq z_N), \quad \text{(A2)}$$

where $p^{(N+1)}(z_N)$ and $s^{(N+1)}(z_N)$ denote the *P*- and *S*-wave potentials at the interface on the top of the half-space. Modes on the (+, −) sheet should satisfy $\mathrm{Re}\{\gamma^{(N+1)}\} > 0$ and $\mathrm{Re}\{\nu^{(N+1)}\} < 0$. Then we can easily realize that the *P*-wave potential decreases with depth and the *S*-wave potential increases with depth in the half-space. Because the OP modes are mainly composed of *S* waves, the growth of *S* waves is more notable in their eigenfunctions.

It seems that the leaky-mode eigenfunctions are inconsistent with the physical wavefields that should vanish at infinity. This conflict can be explained by retracing to Section 2.1 where we mentioned that the leaky waves are the sum of the contributions of leaky modes calculated by the residual theorem and the integrals along the vertical branch cuts (Fig. 1b), which in effect is equal to the integral along the Sommerfeld cuts (Fig. 1a) in mathematics. The Sommerfeld-cut integral, undoubtedly, vanishes at infinity because the integral path is always on the (+, +) sheet. In other words, the leaky modes and the (vertical) branch-cut integrals decompose the leaky waves but leaky modes themselves are not required to satisfy the radiation boundary condition. For leaky waves inside the waveguide, leaky modes can well describe the wave behaviours and the loss mechanisms, which makes them very appealing.

APPENDIX B: THE GROUP VELOCITIES OF PL MODES

Due to the complex wavenumbers of leaky modes, the group velocity is defined as:

$$v_g = \frac{\partial\omega}{\partial \mathrm{Re}\,(k)}. \qquad \text{(B1)}$$

Numerically, the group velocity for a given mode can be quickly estimated by the following finite-difference approach:

$$v_g \approx \frac{(\omega + \Delta\omega) - \omega}{\mathrm{Re}\,[k_n(\omega + \Delta\omega)] - \mathrm{Re}\,[k_n(\omega)]}, \qquad \text{(B2)}$$

where $k_n(\omega)$ denotes the eigen wavenumber of a given mode.

For the second PL mode in Fig. 15, the phase- and group-velocity curves are shown in Fig. B1(a). The group velocities become quite large where the slope of the phase-velocity dispersion curve turns positive. At the same time, the attenuation coefficients increase as well (Fig. B1b). The amplitude ratio between the *P*- and *S*-wave potentials indicates that the PL modes exhibiting high group velocities contains more *S* waves, which is consistent with their high attenuation coefficients because the modes on (+, −) sheet leak *S* waves into the half-space.

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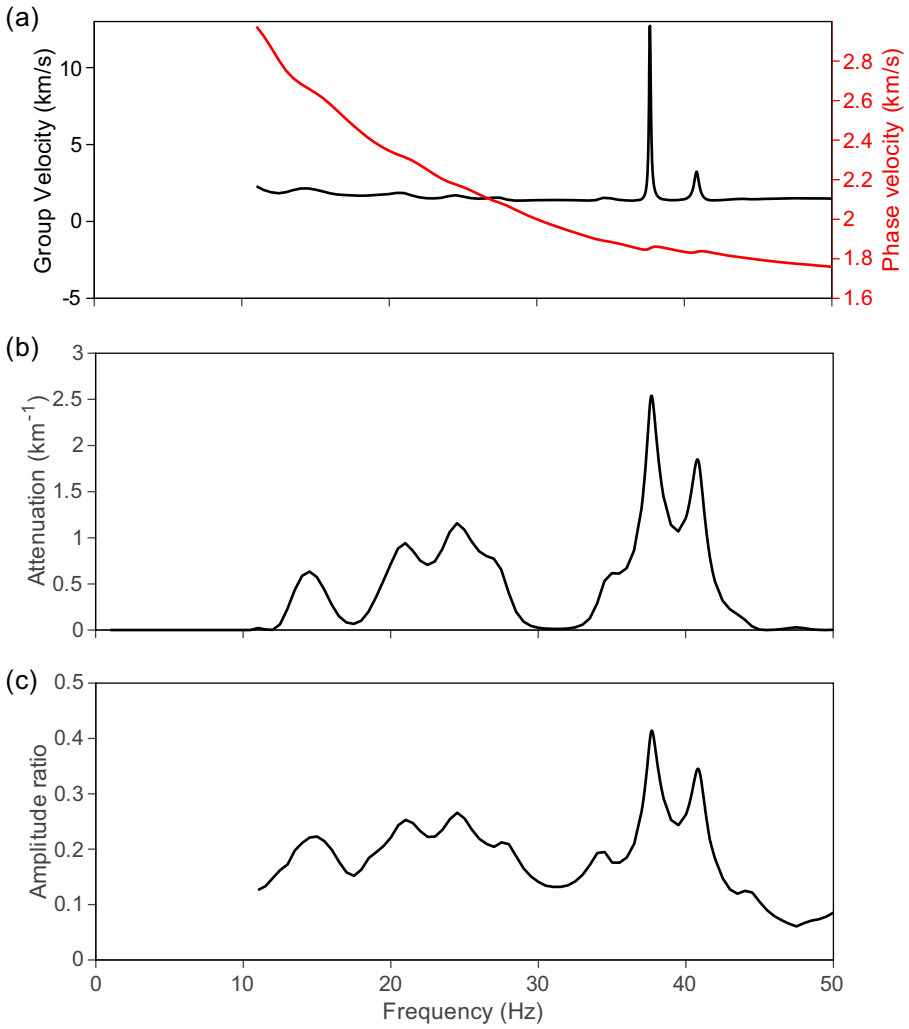


Figure B1. Dispersion of the second PL mode in Fig. 15: (a) phase velocity (red line) and group (black line) velocity versus frequency, (b) attenuation curve and (c) amplitude ratio between the *S*- and *P*-wave potentials within the waveguide.

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